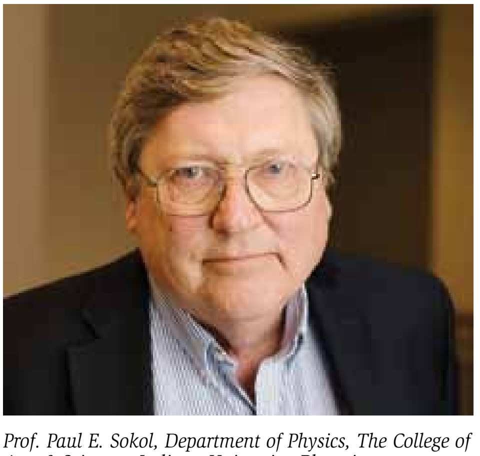
RESEARCH & INNOVATION UPDATES

Tiny artificial luminescent muscles developed that let flying bots to also converse

Taking inspiration from nature, MIT researchers have created electro-luminescent soft artificial muscles that enable flying, insect-scale robots. When the robots fly, these tiny artificial muscles that drive their wings generate a coloured light. The robots can also converse with one another because of this electro-luminescence. For example, robots may use lights to signal others and call for assistance if they were deployed on a search and rescue operation inside a collapsed building. The artificial muscles that these robots have are durable actuators that are made by alternating ultra-thin layers of elastomer and carbon nanotube electrodes in a stack and then rolling it into a squishy cylinder.

Inspired by fireflies, MIT researchers have created soft actuators that can emit light in different colours or patterns (Credit: MIT)



Prof. Paul E. Sokol, Department of Physics, The College of Arts & Sciences, Indiana University, Bloomington (Credit: Indiana University)

Researchers are exploring behaviour of particles in shrinking microchip leads

The wires that transport electricity to microchips must advance in size, speed, and functionality at the same rate as the microchips do. But under newly designed systems, shrinking wires is like confining the electrons in a one-dimensional tube, and that behaviour is quite different from that of a regular wire. Researchers at Indiana University and the University of Tennessee have now built a model system of electronics packed inside a one-dimensional tube to explore the behaviour of particles in these conditions, with the help of an unusual element—helium. Helium was chosen as a model system for their research because of its well-known interactions with electrons and the ease with which it can be made incredibly pure.